Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/31/2016
PI Name:	Cooper, Maya M.S.		
Project Title:	Effects of Processing and Subsequent Storage on Nutrition		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2008	**End Date:**	10/01/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Douglas, Grace	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2015 for reporting purposes (Ed., 10/27/15)		
Key Personnel Changes/Previous PI:	(Sept 2014 report): No personnel changes within the last year. NOTE: Maya Cooper took over as PI, replacing Michele Perchonok, as of 10/1/2008. See Perchonok for FY2008 record. [Information per M. Perchonok/JSC]		
COI Name (Institution):	Douglas, Grace (NASA)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
	NASA has established the goal of returning human expeditions to the moon and extending exploration to Mars. The length of proposed Mars missions increases the importance of nutritional content in the food system. Proper nutrition is essential to prevent or mitigate some physiological effects of long-term space travel as well as maintain astronaut health. The purpose of this task is to evaluate the nutritional content of the existing International Space Station (ISS) and Shuttle food system for a mission to Mars by determining the effects of thermal processing, freeze drying, irradiation, and subsequent storage time on the nutritional content of the food. In order to insure the safety of the food system and in order to deactivate metabolic enzymes that may cause the food to spoil, the packaged foods are processed to commercial sterility. This level of processing can reduce the quality of the food including nutritional content and		

Task Description:	acceptability. There has been limited measurement of the nutritional content of the flight food items. Some macronutrients and minerals are determined chemically at the Johnson Space Center Water and Food Analytical Laboratory (WAFAL). However nutrients, such as vitamins, are currently only determined through a computer program, Food Processor (ESHA Research), which calculates the combined nutritional content based on the ingredients used in the food product's formulation. The computer program provides only a close approximation of the true nutrient content and does not take into account the loss of nutrients during the thermal, freeze drying, or irradiation processes. While kinetic data have been presented in the literature for the loss of nutrition during processing, the kinetic constants provided are specific to the food and the conditions in each test. Therefore, use of the data and kinetic models in the literature will only provide a rough estimate of remaining nutritional content. Fifteen to twenty-six processed food items will be selected per year for five years and an entire nutritional profile will be determined by an off-site laboratory on each of these products at three points of its shelf life – one month after processing, at 1 year, and at 3 years - until nutritional adequacy or a need for countermeasures is established. The one year time period was chosen because it is currently the average time after production that these food items are consumed on ISS and Shuttle missions. Food items will be selected for testing based on the importance of assumed nutritional content, frequency in the menus, and the type of food. Products will be stored at ambient conditions during the 3 year period to mimic the environmental storage conditions of the flight food. This study does not seek to address the effect of processing on nutrients in detail, but rather aims to place in context the overall nutritional status at the time of consumption. Data showing the effect of processing and subsequent storage will be of value pending the need to develop an "Adequate Food System" or institute countermeasures to provide health promoting foods for long duration manned space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	NASA Human Research Program research to determine the nutrient content of prepackaged space food over time will be the most comprehensive look to-date at the extent to which vitamins degrade in complex food dishes over multiple year periods. Manned space missions beyond low-Earth orbit require a food system that can maintain food quality for up to five years. The food should not only be acceptable in taste but retain much of the starting concentrations of vitamins. This study of 109 foods and 25 micronutrients over a 3-year product life is helping to identify the food factors that promote and/or inhibit vitamin sustainment in shelf stable food. The research has implications to the selection and formulation of relief food supplies and military rations that are commonly stored at ambient conditions for long periods before use. Populations that require these food stores generally have limited access to other food so it is desired that the food be as nutritious as possible.
Task Progress:	The task was completed in September 2015. This study evaluated the nutritional suitability of the existing food items, utilized by current NASA programs, for long duration space exploration. The data indicate that several nutrients are nutritionally inadequate or degrade to inadequate levels in the food system over the shelf life required for a Mars mission, and mitigation strategies or countermeasures will be required. From 2008 – 2012, a total of 109 space foods were collected directly after processing and packaging and sent to a subcontracted laboratory to determine the post-stabilization content of 23 micronutrients. At the same time, duplicate samples of the food were placed in a 21°C environmental chamber for storage. After one year of storage and three years of storage passed, these duplicate samples were pulled from the chamber and sent to the subcontracted laboratory for repeat nutritional analysis. Initial, one-year, and three-year nutritional analysis was gathered for each food. Overall, the processing impacts of irradiation, thermostabilization, and freeze-drying on the nutritional quality of space food were aligned with previous researcher findings. Vitamin stability was dependent upon the micronutrient, the food matrix, and the processing method. Although in most foods vitamin C decreased during stabilization processing, there were instances where vitamin C appears to be sustained. Vitamin A, folic acid, thiamin, niacin, and vitamin B6 concentrations are dependent upon the food matrix and the processing method. Only four micronutrients – potassium, calcium, vitamin K, and vitamin D - have inadequate levels in the overall diet initially. The levels are not due to degradation but to low formulated levels in the space food system. Vitamin D is already supplemented but countermeasures for the other micronutrients should be considered. The storage data from the study would suggest that significant degradation occurs with vitamin A, vitamin C, thiamin, and vitamin B6. More data would likely be needed to assess the stability of folic acid and vitamin B12. Degradation also occurs with other vitamins on a more food-specific basis but without substantially shifting the nutritional delivery of the space food system. Because of the prevalence of the vitamins within many foods and high initial concentrations, only the degradation of thiamin and vitamin C across three years of ambient storage are expected to have significant impact to nutritional delivery within our period of interest if degradation rates hold consistent after year three. Thiamin delivery is projected to be inadequate after the first year of storage. Deficiency in even one nutrient could lead to crewmember illness or death. To reduce the risk that crewmember food choice impacts the total nutrition delivery, an effort to stabilize the nutrients of the degrading foods should be considered. Nutrition countermeasures could include changing the processing method to protect nutrients initially, fortifying foods with additional nutrient supply, or reformulating with the more stable food matrices. The possibility also exists for cold storage of the fresh space food supply to suspend much of the vitamin degradation that occurs. Validation of nutritional quality and organoleptic acceptability of any of these changes must occur before implementation.
Bibliography Type:	Description: (Last Updated: 04/23/2019)
Articles in Peer-reviewed Journals	Cooper M, Perchonok M, Douglas GL. "Initial assessment of the nutritional quality of the space food system over three years of ambient storage." NPJ Microgravity. 2017 Jun 9;3:17. eCollection 2017. https://doi.org/10.1038/s41526-017-0022-z ; PubMed PMID: 28649639; PubMed Central PMCID: PMC5466603 , Jun-2017